

My MLE would say it's more likely that the data came from $\text{Ber}(p) = 0.2$ rather than $\text{Ber}(p) = 1$.

I will now do the same for $p \in [0, 1]$

And check which one was actually giving me this data

Succinct:

Set of Ber outcomes n : $\underbrace{1, 1, 1, 0, \dots, 0}_n$

STEP I:

Step 1 = Check $(x_1 = 1, x_2 = 1, x_3 = \dots, x_n = 1)$

STEP II:

Now MLE would compute the entire range $p \in (0, 1)$ and check which one is more likely of occurring of actually giving me this data

EXAMPLE:

likelihood of Bernoulli trial:

Bernoulli trials. If $x_1, \dots, x_n \stackrel{iid}{\sim} \text{Ber}(p)$ for $p \in (0,1)$.

$$\triangleright \mathcal{E} = \{0,1\}^n$$

$$\theta = (0,1)$$

$$\forall (x_1, \dots, x_n) \in \{0,1\}^n, \forall p \in (0,1)$$

$$L(x_1, \dots, x_n, p) =$$

$$\text{Sol}^n: L(x_1, \dots, x_n, p) = \prod_{i=1}^n P_p[X_i = x_i]$$

Lower case $x_i \in$ (observations you see) $= \prod_{i=1}^n P_p[X_i = x_i]$

I want to compute likelihood for this model

In this class, we will always use this notation

We can write this as:

↓
(since we don't know)

$$\rightarrow X_i = x_i = \begin{cases} 1 \\ 0 \end{cases} = p \\ = p \text{ or } 1-p.$$

$$= \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i}.$$

$$\begin{aligned} \underline{\underline{\text{Ans:}}} &= \left(p^{x_i} (1-p)^{1-x_i} \right) \times \left(p^{x_i} (1-p)^{1-x_i} \right) \dots \quad n \text{ times.} \\ &= p^{\sum_{i=1}^n x_i} \times (1-p)^{n - \sum_{i=1}^n x_i} \end{aligned}$$

NOTE: Every time we get data, I will try
to maximize it.

Let $(\mathcal{E}, \{p_\theta\}_{\theta \in \Theta})$ denote a discrete statistical model. p_θ denote pmf p_θ .

$x_1, \dots, x_n \stackrel{\text{iid}}{\sim} p_{\theta^*}$. p_{θ^*} is unknown.

Likelihood fn. is.

$$L_n: \mathcal{E}^n \times \Theta \rightarrow \mathbb{R}.$$
$$(x_1, \dots, x_n, \theta) \mapsto \prod_{i=1}^n p_\theta(x_i).$$

For our purposes, $x_1, \dots, x_n$ denote observations of $X_1, \dots, X_n$.

$$p(x) \ln \left(\frac{p(a)}{p(b)} \right)$$

$$p(a) \ln \left(p(a) - p(b) \right).$$

$$p(a) \ln 1$$

Recitation 4:

Questions: 1.

- ① For $p, q \in (0, 1)$ and $n \geq 1$, compute the KL divergence b/w $B(n, p)$ and $B(n, q)$.

Solⁿ:

$$\text{KL divergence} = \sum p(x) (\log(p(x)) - \log(q(x)))$$

$$\Rightarrow \text{Bin}(n, p) = \sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r}$$

$$= \sum_p p(p) \times \log \left(\frac{p(p)}{p(q)} \right)$$

$$= \sum_p p(p) \times \log \left(\frac{\binom{n}{r} p^r (1-p)^{n-r}}{\binom{n}{r} q^r (1-q)^{n-r}} \right)$$

$$= np \left(r \log \left(\frac{p}{q} \right) - (n-r) \log \left(\frac{1-p}{1-q} \right) \right) \left(\frac{p}{q} \right)^r \left(\frac{1-p}{1-q} \right)^{n-r}$$

We know: Expectation of $\text{Bin}(n, p) = np$.

$$\Rightarrow \sum_{k=0}^n \text{Bin}(n, p) \times k = np$$

$k \Rightarrow k \text{ known}$

$$\therefore = \left(\sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} \right) \log \left(\frac{p}{1-p} \right) + \left(\sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} \right) \log \left(\frac{1-p}{p} \right)$$

$$= np \log \left(\frac{p}{1-p} \right) + (n-np) \log \left(\frac{1-p}{p} \right)$$

Things to observe:

$$\Rightarrow h_p = 0, \quad p \in [0, 1] : K_2(p, q) = \infty.$$

Exercise:

$$q \rightarrow 1; \quad p \rightarrow 0; \quad p \rightarrow 1,$$

② Compute KL divergence between 2 Gaussian distributions, that have the same variance.

$$P = \mathcal{N}(a, 1), Q = \mathcal{N}(b, 1) \quad a, b \in \mathbb{R};$$

$$f_{a, 1}(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-a)^2}{2}}$$

$$KL(P||Q) = \int_{\mathbb{R}} f_{a, 1}(x) \ln \left[\frac{f_{a, 1}(x)}{f_{b, 1}(x)} \right] dx$$

$$= \int f_{a, 1}(x) \times \ln \left[\frac{e^{-\frac{(x-a)^2}{2}}}{e^{-\frac{(x-b)^2}{2}}} \right] dx$$

$$= \int f_{a, 1}(x) \times \ln \left[e^{\frac{(x-b)^2}{2} - \frac{(x-a)^2}{2}} \right] dx$$

$$= \int f_{a, 1}(x) \left(\frac{(x-b)^2}{2} - \frac{(x-a)^2}{2} \right) dx$$

$$= \int f_{a, 1}(x) \left[x(a-b) - \frac{a^2}{2} + \frac{b^2}{2} \right] dx$$

$$= (a-b) \int x f_{a, 1}(x) dx + \left(-\frac{a^2}{2} + \frac{b^2}{2} \right) \int_{\mathbb{R}} f_{a, 1}(x) dx$$

$$(a-b)a + \frac{b^2 - a^2}{2} = a^2 - ba + \frac{b^2}{2} - \frac{a^2}{2} = \frac{a^2 + b^2 - 2ab}{2} = \frac{1}{2}(a-b)^2$$

Q). Show that the Poisson Distribution with parameter $\frac{1}{n}$ converges in total variation to the Dirac Distribution.

Solⁿ:

$$P_n = \text{Poi}\left(\frac{1}{n}\right) \quad Q = \delta_0. \quad \delta_n = \delta_0. \quad \delta_n(P_n, Q) \xrightarrow{n \rightarrow \infty} 0.$$

$$P_n(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, \dots \quad Q(k) = \begin{cases} 1 & k=0 \\ 0 & \text{otherwise} \end{cases}$$

↓
Dirac.

Proof: $TV(P_n, Q) = \frac{1}{2} \sum_{k=0}^{\infty} |P_n(k) - Q(k)|$

$$= \frac{1}{2} \left[\frac{1}{e^n} + \sum_{k=1}^{\infty} P_n(k) \right]$$

$$\xrightarrow{n \rightarrow \infty} = \frac{1}{2} \left[\frac{1}{e^0} + \sum_{k=1}^{\infty} P_n(k) \right]$$

$$= \frac{1}{2} \left(0 + \frac{1}{2} \left[P_n(1, 2, \dots) \right] \right)$$

$$= \frac{1}{2} \left(1 - \frac{1}{e^n} \right)$$

$$= \frac{1}{2} \left/ 1 - \frac{\left(\frac{1}{n}\right)^0 e^{\frac{1}{n}}}{0!} \right/ \xrightarrow{n \rightarrow \infty} \frac{1-1}{2} = 0$$

Distance between Probability
Distributions:

P, Q, R pd over E , with densities p, q, r
[pmf P, Q, R]

$$A \subseteq E = P(A) = \int_A p(x) dx$$

$$= \sum p_i$$

LECTURE 9: INTRODUCTION TO MLE;

9.2

→ Looking for Distance b/w Probability Distributions.

↓
Try with Total Variation.
↓
No nice estimation

→

↓
Then KL divergence → (Replace expectation by avg)

↓
Minimizing the estimate of KL
↓

Maximum Likelihood → maximization.

↓
fn of (μ, θ)

Bernoulli Trials:

$$h(x_1, \dots, x_n; p) = p^{\sum x_i} \times (1-p)^{n - \sum x_i}$$

$$x_1, \dots, x_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$$

$$f_{\mu, \sigma^2}(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\prod_{i=1}^n f_{\mu, \sigma^2}(x_i) = \frac{1}{\sigma^n (2\pi)^{n/2}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$$

$$\frac{1}{2} \left(\frac{(x_i - \mu)^2}{\sigma^2} \right) = y_i$$

Gaussian:

$$L(x_1, \dots, x_n; \mu^2) = \frac{1}{(\sigma\sqrt{2\pi})^n} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$$

$$e^{-\frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2}$$

$$\sqrt{e^{x^2}}$$

$$e^{x^2} = e^{x^2} \cdot e^{2xy_1}$$

$$\log e^{x^2} = x^2$$

$$x^2 = y$$

$$e^{\frac{(x-\mu)^2}{2\sigma^2}} = e^{\frac{y}{2}}$$

$$\log \left(\frac{(x-\mu)^2}{2\sigma^2} \right) = \frac{y}{2}$$

LIKELIHOOD OF AN EXPONENTIAL DISTRIBUTION:

$$x_1, \dots, x_n \stackrel{iid}{\sim} \text{Exp}(\lambda)$$

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0.$$

$$\prod_{i=1}^n f_X(x_i) = \lambda^n e^{-\lambda \sum_{i=1}^n x_i} \prod_{i=1}^n \mathbb{I}(x_i > 0)$$

$$\begin{aligned} E &\rightarrow (0, \infty) \\ \partial &\rightarrow [0, \infty) \end{aligned}$$

$$\prod_{i=1}^n P(x_i > 0) = \begin{cases} 1 & \text{if } x_1 > 0 \text{ \& } x_2 > 0 \dots \\ 0 & \text{otherwise} \end{cases}$$

The reason I don't care is those are not US || $w(x_i > 0) -$

$\Rightarrow$ If this indicator does not depend on the parameter, whether I put it or not, it's not going to matter.

Those are actually my observations

They are well specified models.



They come from an exponential distribution, with parameter, which means they are all +ve,



They are well specified which means



They come from an exponential distribution.



If those guys are larger than λ .



Then I have to keep track of it.

$$\therefore L(x_1, \dots, x_n, \lambda) = \lambda^n e^{-\sum_{i=1}^n x_i} I(\min x_i > 0)$$

$$f_{\lambda}(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

$$\prod_{i=1}^n f_{\lambda}(x_i) = \lambda^n e^{-\lambda \sum_{i=1}^n x_i} \prod_{i=1}^n \mathbb{1}_{x_i \geq 0},$$

$$\prod_{i=1}^n \mathbb{1}_{x_i \geq 0} = \begin{cases} 1 & \text{if } x_1 \geq 0 \text{ \& } x_2 \geq 0 \dots \\ 0 & \text{otherwise.} \end{cases}$$

$$= \prod_i \mathbb{1}_{x_i \geq 0}$$

→ these nos. are already ≥ 0
so no need of an indicator here

$$\therefore L(x_1, x_2, \dots, x_n, \lambda) = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

(don't use

the indicator here because it does not depend on λ).

$$= \lambda^n e^{-\lambda \sum_{i=1}^n x_i} \prod_i \mathbb{1}_{x_i \geq 0}$$

LIKELIHOOD OF A UNIFORM DISTRIBUTION.

$$x_1, \dots, x_n \sim U([0, b]),$$

↑
two ways U or Unif.

$$\rightarrow \mathcal{E}: [0, \infty)$$

$$\rightarrow \theta: [0, \infty).$$

$$f_b(x) = \frac{1}{b} \mathbb{I}(0 \leq x \leq b).$$

$$\prod_{i=1}^n f_b(x_i) = \frac{1}{b^n} \prod_{i=1}^n \mathbb{I}(0 \leq x_i \leq b)$$

$$= \frac{1}{b^n} \prod_{i=1}^n \mathbb{I}_{\max\{x_i, 0\} \leq b} \times \mathbb{I}_{\max\{x_i, 0\} \leq b}.$$

↑
($\bar{x}$ is max here).

$$\therefore L(x_1, \dots, x_n, b) = \frac{1}{b^n} \mathbb{I}_{\max x_i \leq b}.$$

→ For $U[0, b) \rightarrow$ You estimate the data, and there is -25 in there. →

Issue with data-set, & Issue with model.

N. is well specified.

RULE: Indicator does not depend
on the unknown parameter.

$f(x)$ be a continuous r.v. fn. with $\max f(x) \leq 1$.
 $0 < f(x) \leq 1$.

Which of the following fns have the same
 $\arg\max$ as $f(x)$?

Arguments of the minima (resp arguments
of the maxima) of a fn $f(x)$, denoted by $\arg\min f(x)$ (resp
 $\arg\max f(x)$), is the VALUES OF x at which,
 $f(x)$ is minimum (resp (maximum)).

We can also, restrict ourselves to
a subset S of the domain of f , and denote
by $\arg\min_{x \in S} f(x)$ (resp. $\arg\max_{x \in S} f(x)$), the
values at which $f(x)$ is min^m, resp. max over S .

$\max_x f(x) = 1$ (Note that $\max_x f(x)$ is the maximum value of the fn., which is different from $\arg \max_x f(x)$, the value of the argument x at which the fn. is max.)

Which of the following fns. of $f(x)$ has the same $\arg \max$ as $f(x)$.

Maximum Likelihood Estimator.

Let $X_1, \dots, X_n$ be an iid sample with a statistical model $(\mathcal{F}, (p_\theta)_{\theta \in \Theta})$ and let L be the corr. likelihood.

Definition.

MAXIMUM LIKELIHOOD ESTIMATOR of θ is defined as:

$$\hat{\theta}_n^{MLE} = \arg \max_{\theta \in \Theta} L(X_1, \dots, X_n, \theta)$$

provided it exists.

Remark / Log-Likelihood Estimator:

In practice, we use the fact that

$$\hat{\theta}_n^{MLE} = \arg \max_{\theta \in \Theta} \ln L(X_1, \dots, X_n, \theta)$$

If some fn.

$$\max_{\theta \in \Theta} f(\theta) = f(\theta^*). \Leftrightarrow \theta^* = \arg \max_{\theta \in \Theta} f(\theta)$$

Maximizing a fn. or maximizing an increasing fn. of this +ve fn.; for ex,



It turns out, if I maximize an increasing fn. of this fn. it's the same thing, at least

in terms of $\arg \max$, if I take its log, which is an increasing fn. over the +ve nos.

Remark:

→ When we took pdf/pmf, I index them by the parameter. Argument $\rightarrow x$.

↓ kept p fixed, thought it as $f(x)$.

But IT'S a fn. of θ and p

once log-likelihood is written, we're going to maximize it in the parameter. We think of it as a fn of μ, σ^2, λ etc.

INTUITION:

Think in terms of pdf/pmf. as fn. of your data.



THINK OF IT AS A FN. OF your parameter (θ, x) .



Then we have to do this max. likelihood,
 $\nwarrow$
 how to maximize!

CONCAVITY IN ONE DIMENSION:

OPTIMIZATION:

→ Maximizing or minimizing a fn. is referred to as optimization.



we will focus on maximization.



$$\min_{\theta} - \ln(\theta) \Leftrightarrow \max_{\theta \in \Theta} \ln(\theta).$$

→ ML Problems don't want to directly deal with likelihood → that's what we talk about in M-estimation.

Problem in like $x \in \mathbb{R}^{20,000}$.

→ This is exactly the same problem:
↓

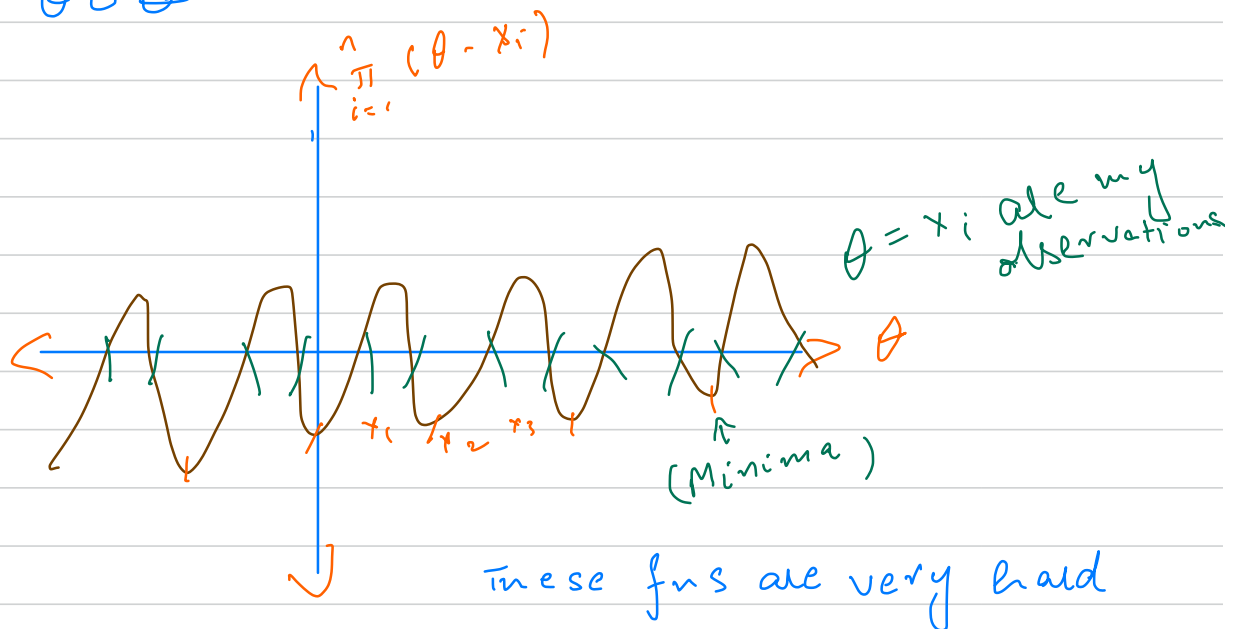
Maximizing means minimizing negative of function.

↓
Again when it's coming to argmax of the argument

↑ This is minimization

Note that

$$\min_{\theta \in \Theta} -h(\theta) \iff \max_{\theta \in \Theta} h(\theta)$$



to minimize

CONCAVE AND CONVEX FUNCTIONS:

Defⁿ:

CONCAVE: A fn. twice differentiable $h: \Theta \subset \mathbb{R} \rightarrow \mathbb{R}$.
if its second order derivative satisfies

$$h''(\theta) \leq 0. \quad \forall \theta \in \Theta.$$

It is strictly concave $h''(\theta) < 0$.

(STRICTLY) CONVEX if:

- $h(\theta)$ is (strictly) convex i.e. $h''(\theta) \geq 0$ (or $h''(\theta) > 0$)
CONCAVE;

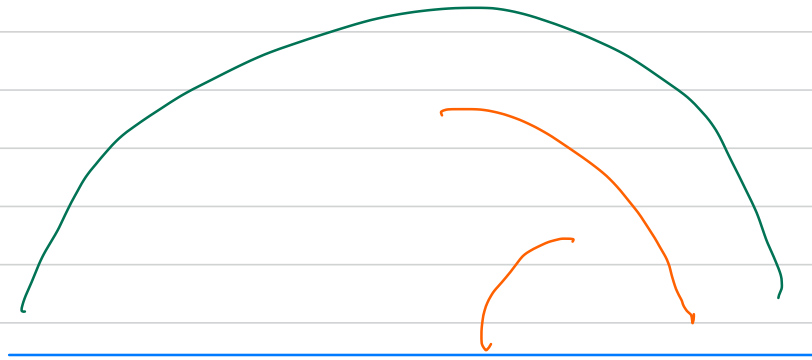
$$\begin{aligned} \Theta &= \mathbb{R}, & h(\theta) &= -\theta^2 \\ \Theta &= (0, \infty), & h(\theta) &= \sqrt{\theta} \\ \Theta &= (0, \infty), & h(\theta) &= \log(\theta) \\ \Theta &= [0, \pi], & h(\theta) &= \sin(\theta) \\ \Theta &= \mathbb{R}, & h(\theta) &= 2\theta - 3 \end{aligned}$$

$$-4x^3 + 2x - 40.$$

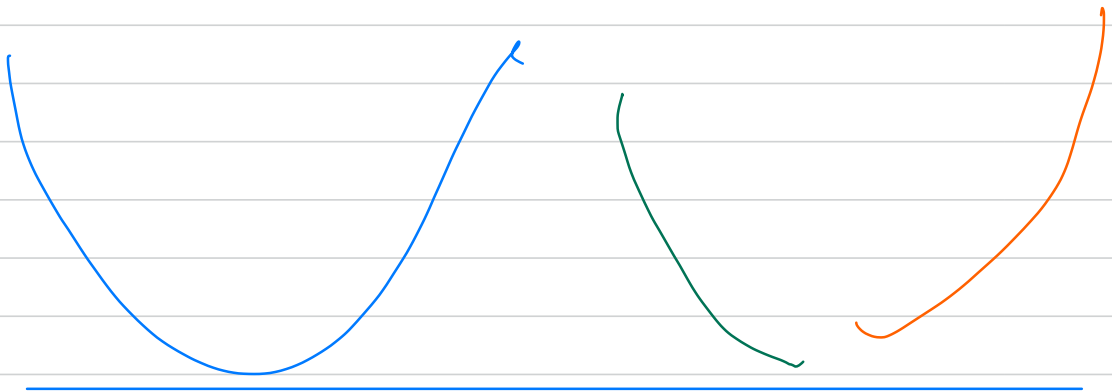
$$-6x^2 + 2.$$

$$\begin{pmatrix} -4x + 2 \\ 2x - 5 \end{pmatrix} = 0.$$

Concave fns.



CONVEX FN EXAMPLES



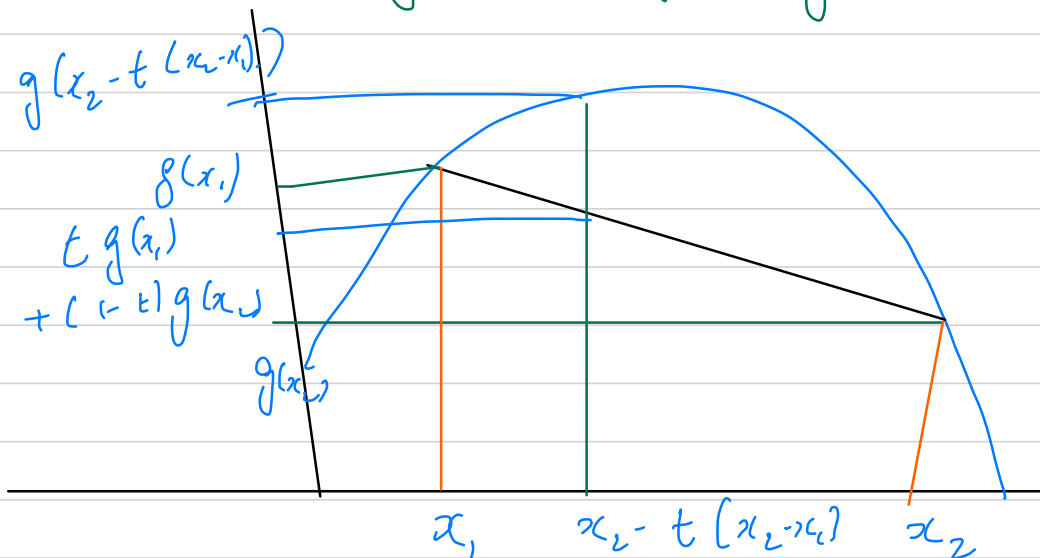
CONCAVITY IN ONE DIMENSION:

A fn $g: I \rightarrow \mathbb{R}$ is concave (or concave down), where I is an interval if for all pairs of real numbers,

$$x_1 < x_2 \in I$$

$$g(tx_1 + (1-t)x_2) \geq tg(x_1) + (1-t)g(x_2)$$

Geometrically, this means that for $x_1 < x < x_2$, the graph of g is above the secant line connecting the 2 points $(x_1, g(x_1))$ and $(x_2, g(x_2))$.



At $x = x_2 - t(x_2 - x_1) = tx_1 + (1-t)x_2$ the y -value of the graph g is $g(x) = g(tx_1 + (1-t)x_2)$ while

the y-value of the secant line is

$$tg(x_1) + (1-t)g(x_2).$$

If the previous inequality is strict, i.e. if:

$$g(tx_1 + (1-t)x_2) > tg(x_1) + (1-t)g(x_2) \quad \forall 0 < t < 1.$$

then g is STRICTLY CONCAVE.

VICE-VERSA FOR CONVEX.

If in addition g is twice differentiable in the interval I , i.e., $g''(x)$ exists $\forall x \in I$ then g is

CONCAVE: IF AND ONLY IF $g''(x) \leq 0 \quad \forall x \in I$.

STRICTLY CONCAVE: IF $g''(x) < 0 \quad \forall x \in I$.

CONVEX if and only if $g''(x) \geq 0 \quad \forall x \in I$.

STRICTLY CONVEX if $g''(x) > 0 \quad \forall x \in I$.

(8) Concavity in Higher Dimensions: Gradient, Hessians, Semi-Definiteness.

$\rightarrow N(\mu, \sigma^2) \text{ MGR, } \sigma^2 > 0.$

Now, we are going to discuss it in
subset $\mathbb{R}^2$.

$$= \frac{1}{2} \int_0^1 e^{-x} - 1 + \int_1^{\infty} e^{-x} dx.$$

$$= \frac{1}{2} \left[\int_0^1 (e^{-x} - 1) dx + \int_1^{\infty} e^{-x} dx \right]$$

$$= \frac{1}{2} \left[\left[-e^{-x} \right]_0^1 - 1 + \left[-e^{-x} \right]_1^{\infty} \right]$$

$$= \frac{1}{2} \left[\left(-\frac{1}{e} - 1 \right) + \left(0 - \left(-\frac{1}{e} \right) \right) \right]$$

$$= \frac{1}{2} \left[1 - \frac{1}{e} - 1 + \frac{1}{e} \right]$$

$$= \frac{1}{2} \left[\frac{2}{e} \right] = \frac{1}{e}.$$

Concavity in Higher Dimensions:

$$\frac{\partial}{\partial \theta_1} f(\theta) = f'(\theta).$$

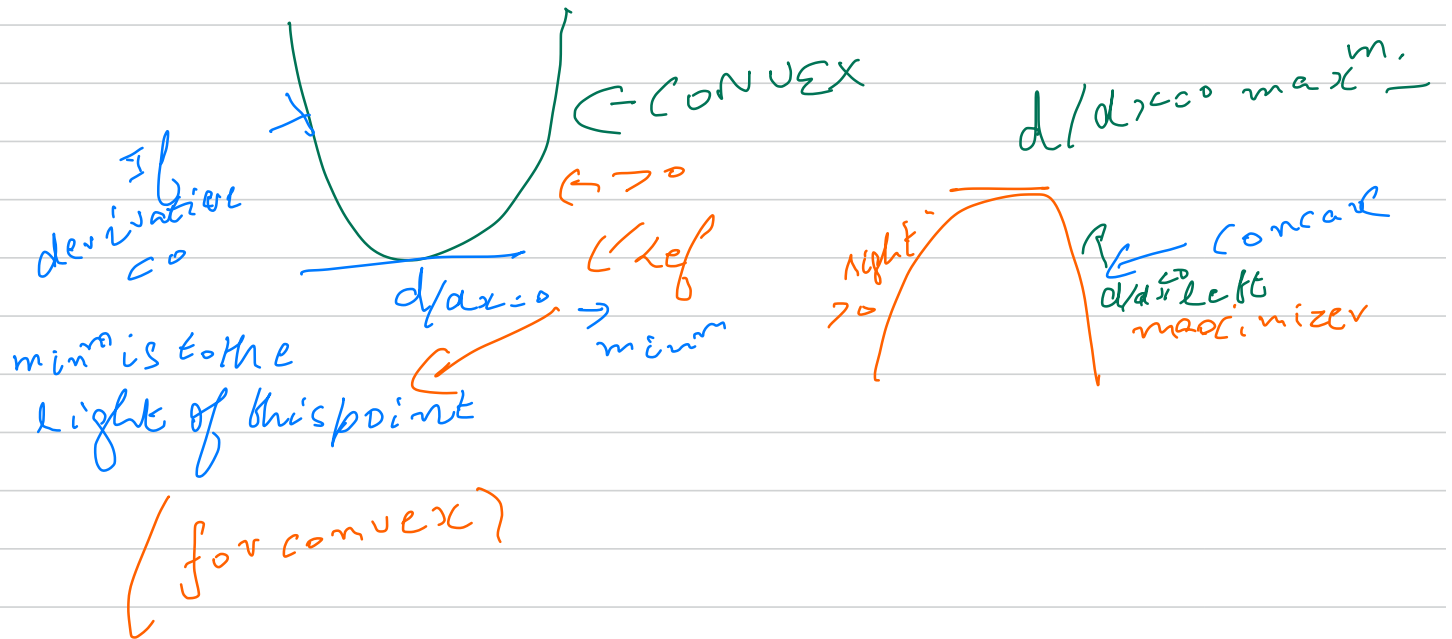
$$\nabla f(\theta) = \begin{pmatrix} \frac{\partial f}{\partial \theta_1} \\ \vdots \\ \frac{\partial f}{\partial \theta_d} \end{pmatrix}$$

I st derivative-

II derivative:

STRICTLY CONCAVE FNS

AND UNIQUE MAXIMIZER;



[IMP].

KKT CONDITIONS:

We set up this optimization problem from a minimization perspective using log-likelihoods:

$$\min_{n_1, \dots, n_6} - \sum_{i=1}^6 k_i \ln(n_i) .$$

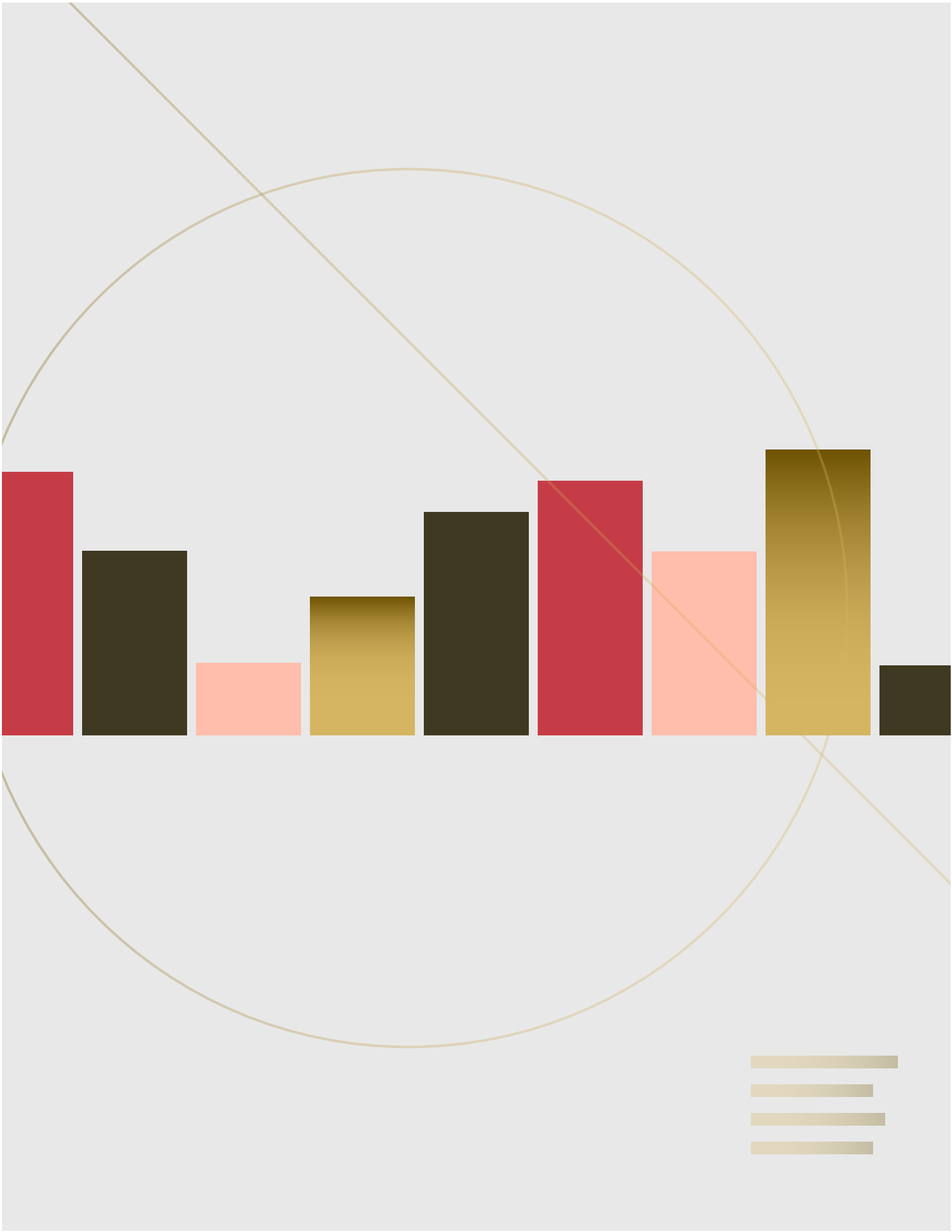
$$\text{Constraints: } \sum_{i=1}^6 n_i = 1 \quad n_i \geq 0 \quad i=1, 2, \dots, 6.$$

Additional notations:

$$\min_{n_1, \dots, n_6} f_0(n_1, \dots, n_6) \triangleq - \sum_{k=1}^6 k_i \ln(n_i) .$$

$$\text{Constraints: } h(n_1, \dots, n_6) \triangleq \sum_{i=1}^6 n_i - 1 = 0 .$$

$$f_i(n_i) \triangleq -n_i \leq 0, \quad i=1, \dots, 6$$



Lecture 10:

→ When is it the case you actually compute without setting a derivative equal to 0?

$x_1, \dots, x_n \stackrel{\text{iid}}{\sim} U[0, \lambda], \lambda > 0$

$$\text{pdf: } f_\lambda(x) = \frac{1}{\lambda} \mathbb{I}_{0 \leq x \leq \lambda}$$

↓ when computing MLE:

In a uniform?

$$L(x_1, \dots, x_n, \lambda) = \frac{1}{\lambda^n} \pi \mathbb{I}_{(0 \leq \min(x_i) \leq \max(x_i) \leq \lambda)}$$

↑
 $\mathbb{I}$ is $\mathbb{I}$ can remove, no matter what set of endpoints I get. because it does not depend on lambda.
 → irrelevant + we

→ In this case, the regularity conditions will always be satisfied for, by computing the maximum log likelihood and, setting their derivative equal to zero.

Now, here, I want to take the log and set it equal to zero.

$$L(x_1, \dots, x_n, \lambda) = \frac{1}{n} \ln \underbrace{\pi \mathbb{I}_{\max x_i \leq \lambda}}_{\downarrow}$$

$$\frac{1}{(\max x_i)^n}$$

If we plot this as a fn. of λ ,

as λ becomes smaller

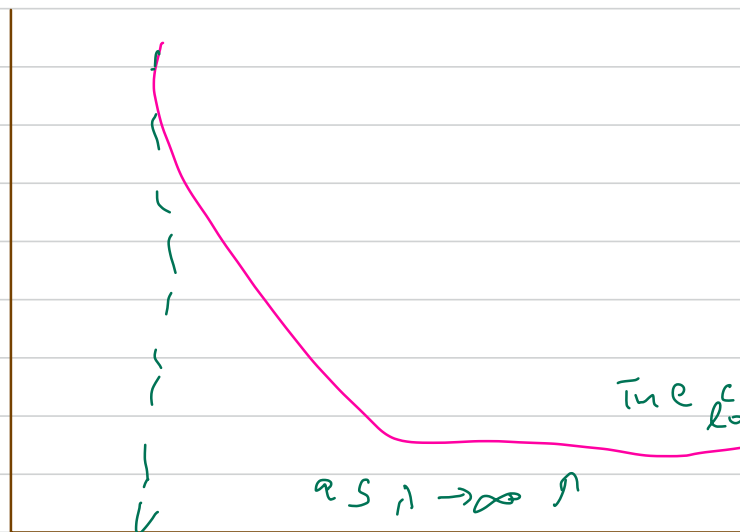
$$\max x_i \rightarrow 0$$

$$\text{coz } 0 \in [0, \lambda] \quad \lambda \rightarrow 0.$$

(does)

$$\max x_i$$

$$\lambda$$



$$\max x_i$$

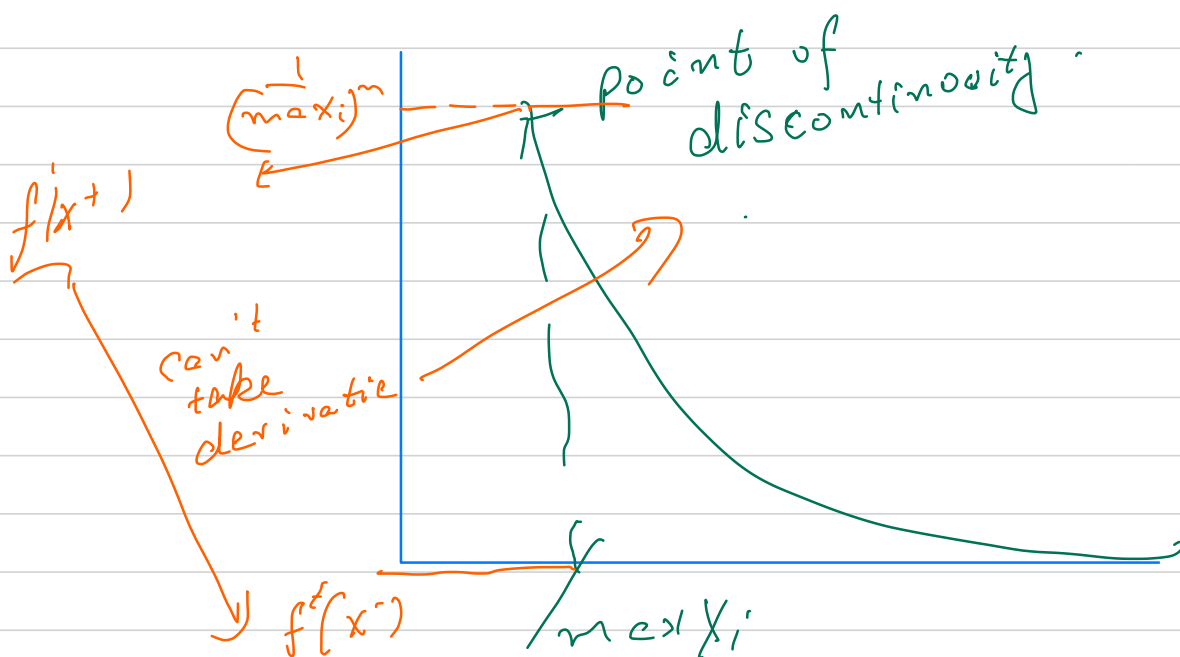
The curve looks like this

$$\text{as } \lambda \rightarrow \infty$$

$$\frac{\partial}{\partial x_i} (\max x_i) = - \quad \text{or} \quad \frac{\partial}{\partial x_i} (\min x_i) = 0$$

claim: (if we want), this is true,

ONLY if this has derivatives everywhere



It's not only not differentiable,
it's not continuous:-

INTUITION: If we plot this fn, look at
at it, we can see where the derivative
is realized

$$\therefore \hat{\lambda}^{MLE} = ?$$

$$\text{Ans: } \hat{\lambda}^{MLE} = \max_i x_i$$

- This model does not satisfy the regularity conditions.
- But still a consistent estimator.
- So this slide will be true, even if the model is not regular.

→ When we talk about asymptotic normality,

we will need
regularity conditions

Not satisfying these conditions
will not be the case.

Regularity means:

If I take the derivative of my likelihood
& my log likelihood = 0

Get my max likelihood.

If $\frac{\partial}{\partial \theta} \ln L(x, \dots, x_n, \theta) = 0 \rightarrow$

If it's differentiable

① Sometimes $\frac{\partial}{\partial \theta} L(x, \theta) = 0 \rightarrow$ Enough

↓
If $\frac{\partial}{\partial \theta} \ln L(x, \theta) \Rightarrow$ not differentiable

② Think twice, when you are taking the
derivate. → If discontinuous
↓ if you do this, non-sense.

Remark;

Plot it: we have a method to compute estimators.

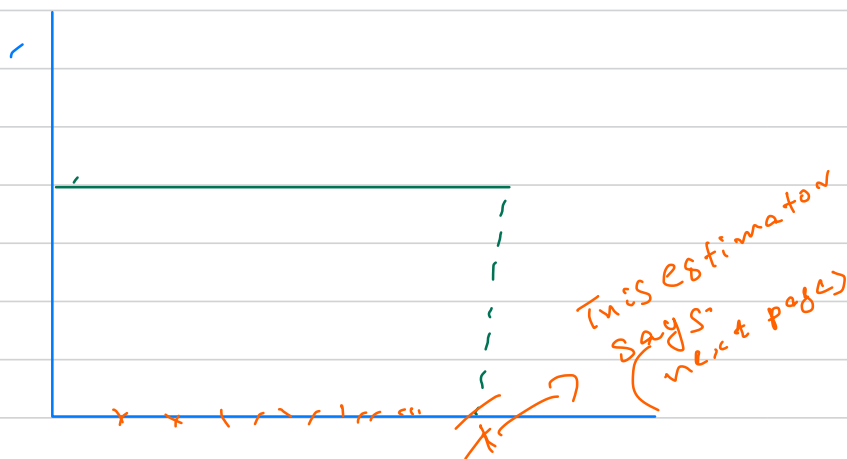
$\downarrow$ $\lambda^{MLE} = \max_i \lambda_i \rightarrow$ Not avg.
 $\cup [0, \lambda] \rightarrow$ Not related to average.

$\downarrow$
up til now $\rightarrow$ take it's expectation, $\frac{1}{2}$ expectation.

$\downarrow$
 $\bar{x}_n$

$\downarrow$
 $\frac{1}{2}$
 $\bar{x}_n$

Remark: If we had done the same,
expectation $\rightarrow \frac{1}{2} \rightarrow 2 \bar{x}_n$



well, the best one should be pretty close to:

↓
I know my estimator

↓
Strictly less than λ .

↙
biased $\Rightarrow$ but a very good estimator,

↓

The true value of my parameter, will
always be larger than, a number

↙
which is what I propose.

↓
ponder: why it actually makes
sense to take this estimator.

→ It's not π_n , or something
coz

Generalization of the loaded die problem:

Question: we observe $k_i, i=1, \dots, 6$, outcomes of result i , out of a total of n rolls of a loaded die problem with probabilities, $\eta_i, i=1, \dots, 6$. How do we obtain ML estimate of $\eta_i, i=1, \dots, 6$.

Solution: $\ln(x_1, \dots, x_n, \eta_1, \dots, \eta_6)$, ignoring the constant terms, can be computed as -

$$\ln(x_1, \dots, x_n, \eta_1, \dots, \eta_6) = \sum_{i=1}^6 (\eta_i)^{k_i}.$$

Finding out of $\eta_i, n, i=1, \dots, 6$, involves maximization of $\ln(x_1, \dots, x_n, \eta_1, \dots, \eta_6) = \sum_{i=1}^6 (\eta_i)^{k_i}$.

Finding out of $\eta_i, i=1, \dots, 6$ involves maximizing $\ln(x_1, \dots, x_n, \eta_1, \dots, \eta_6) = \sum_{i=1}^6 (\eta_i)^{k_i}$ with the following 2 constraints;

$$\sum_{i=1}^6 \eta_i = 1 \text{ and } \eta_i \geq 0, i=1, \dots, 6$$



So ℓ^n known as Karush-Kuhn-Tucker (K-K-T) conditions

MLE:

- 2 ways of computing MLE:

- Either setting derivative to zero.
- Looking at the plot.

Questions:

- ① Is it unbiased.
- ② Does it have small variance
- ③ Does it have a small quadratic risk?
- ④ Is it consistent?
- ⑤ Is it asymptotically normal?

↑
we could say these things when we had $\bar{X}_n$.
↑

Can we still say them?

→ Under mild regularity conditions $\Rightarrow$
↳ Consistent:

$$\hat{\theta}^{MLE} \xrightarrow[n \rightarrow \infty]{P} \theta^*$$

At some point on our way
to compute MLE.

↑
we replaced expectation by
 $\bar{x}_n$.

This is because.

Maximizer of this → something that converges to minimize of this.

→ $\frac{1}{n} \log L(x, \dots, x_n, \theta) \xrightarrow{n \rightarrow \infty} \text{constant}$



KL(P_θ || P_g)

Have this scale → Replace Expectation

by avg.

$\theta = 0$

↑
LLN → AS soon as you do this.

LLN tells you

Avg Converges to Expectation.

↑
(this justified entire process in the first place)

→ $KL(P_{\theta^*} \| P_{\theta}) = 0$
 when $\theta^* = \theta$.

↓

Definite, when the
 2 distributions are same.

↑
 Not just $P_{\theta} = P_{\theta^*}$, $(\theta = \theta^*)$

→ Parameter → Identifiable.

I want this
 to be true.

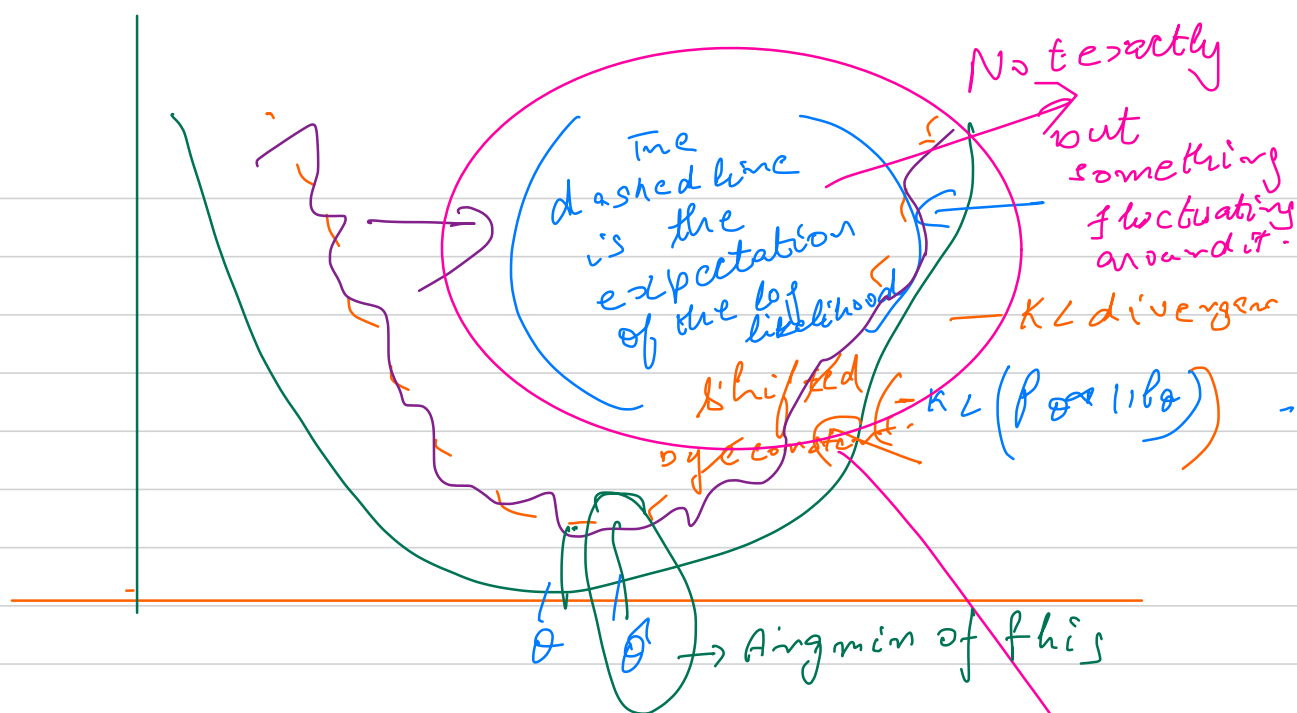
Process:

① Probability Distribution:

↓
 $P_{\theta} \rightarrow \text{Minimizer.}$

↓
 P_{θ^*} Property of KL-
 divergence.

→ $P_{\theta^*} \rightarrow \text{Uniquely identifies } \theta^*$
 ↳ if Parameter
 identifiable.



I have 2 fns $\rightarrow P_{\theta}, P_0$.

Along the y-axis, those 2 things become the same

As $n \rightarrow \infty$
Log likelihood and the dashed line get closer and closer.

$(\hat{\theta} \rightarrow \theta \text{ on the x-axis})$

So, I need to translate y-axis convergence to x-axis convergence

Hopefully this should work.

Technical conditions allow
to transfer this convergence on the y -axis of the
fns $\rightarrow$ To ϵ fn of their minimizers.



This goes wrong when you have
discontinuous fns.



But in general it's actually
fine when you have nice smooth fns!

CONSISTENCY OF MLE:

Covariance, Independence

$$Cov(X, Y) = Cov(Y, X)$$

$$\rightarrow Cov(X, X) = Var(X)$$

If X and Y are independent,

$$\left(\begin{array}{l} \text{If } X \text{ \& } Y \text{ are indep.} \\ \text{endent} \end{array} \right) Cov(X, Y) = 0$$

Conversis true only if

$[C(X, Y)]^T \rightarrow$ Gaussian vector

$\rightarrow$ Being a $\nrightarrow$ Both X and Y are Gaussian

$\left\{ \begin{array}{l} \text{It means any linear combination of} \\ X \text{ and } Y \text{ take remains Gaussian} \end{array} \right\}$

$\rightarrow \alpha X + \beta Y \rightarrow \text{Gaussian } \forall (\alpha, \beta) \in \mathbb{R}^2 \setminus \{0, 0\}$
ie. $\alpha=0, \beta=0$

V. Imp:

Take $X \sim N(0,1)$, $B \sim \text{Ber}(1/2)$, $R = 2B - 1 \sim \text{Rad}(1/2)$

Rademacher vector
i.i.d.

$R = 2B - 1 = 0 \rightarrow \text{If } B = \{1\}$
 $\downarrow$ $-1 \rightarrow \text{If } B = \{0\}$
 linear transformation of Ber.

X, B i.i.d. $\Rightarrow X, R$ i.i.d.

Then

$$Y = RX \sim N(0,1)$$

$$X \sim N(0,1)$$

$$Y \sim N(0,1)$$

$\begin{pmatrix} X \\ Y \end{pmatrix} \rightarrow$ Gaussian vector is not